CDS 535 Data Mining
Group Project Requirement

Team Name: _____
Members and Student ID: _____
Instructor: _____

Deadline: 8:00 pm, 4 November 2026

1 Project Objectives

CDS535 Data Mining Group Project using Weka or You may use other tools, such as Python, as well. Please make sure you can implement our requirements.

- To understand, and apply the fundamental principles and concepts of data science and data mining with hands-on practice.
- To enhance students' understanding of the end-to-end data science process using **Weka** workflows (Explorer/Knowledge Flow/Experimenter).
- To facilitate learning of methods, techniques, and skills in data preparation, analysis, modeling, and mining for real-world problems using filters, classifiers, clusterers, associators, and evaluation tools.

Submission Checklist

- Each group should have 5–6 students.
- Final report (10–12 pages, font size 12) due: **8:00 pm, 4 November 2026**.
- Deliverables:
  - PDF report (Compulsory).
  - Dataset description (Optional).
  - Presentation slides (all members present) — due 9 November 2026.
- File naming: `CDS535_Group_TeamName_Final.pdf`. Include student IDs on the title page.

2 Topics: Data Mining with UCI Machine Learning Repository Datasets

The project topic is **Data Mining with Benchmark Datasets from the UCI Machine Learning Repository**. Each group must select **at least two** datasets from the UCI Machine Learning Repository (https://archive.ics.uci.edu/) (Dua & Graff, 2019) according to their interests, and formulate practical data science problems around **prediction (classification/regression), clustering, or association analysis**. Techniques learned in the course should be properly applied in **Weka** (Frank et al., 2016) (or in Python, if the workflow is fully documented) to solve the problems.

All datasets used must come from UCI. If a group wishes to bring in an additional non-UCI dataset, it must obtain approval from the instructor.

3.1 Dataset Selection Requirements

Each group must satisfy all of the following:

1. At least **two distinct UCI datasets** are analysed, and each dataset must be used for a complete end-to-end pipeline (preprocessing → modeling → evaluation), not merely mentioned.
2. At least one dataset with $\geq$ 1,00 **instances**, so that meaningful features can be extracted and K-fold cross-validation is statistically stable.
3. At least one dataset that contains **missing values or nominal (categorical) attributes**, which forces explicit justification of the preprocessing filters.
4. The two datasets must differ in at least **two** of the following dimensions: size, attribute type, number of classes, missing-value rate, degree of class imbalance. The differences must be exploited: an explicit **comparative analysis** (why does model A win on dataset 1 but lose on dataset 2?) is a required part of the report.
5. Every dataset must be **publicly available and citable**: record the UCI dataset page, the UCI DOI/ID, the original donor, and the license/usage constraints in the References and Data Provenance section.
6. Convert the raw files to **ARFF** (or CSV) and document the schema: number of instances, attributes, types, missing-value coding, and the meaning of each class label.

Practical tip: the standard copies of the UCI benchmarks shipped with Weka (folder `$WEKA_HOME/data`, available in the Explorer's "Open file..." dialog) are accepted. They are the same datasets as the UCI releases, with only minor formatting differences; nonetheless cite the original UCI source. For example, `iris`, `breast-w`, `mushroom` and `heart-statlog` should be attributed to their original donors (Detrano, 1988; Fisher, 1936; Schlimmer, 1987; Wolberg et al., 1994), in addition to the repository itself (Dua & Graff, 2019).

3.2 Recommended UCI Datasets

Tables 1 and 2 list recommended datasets. They are deliberately diverse in size, attribute type, class count and data quality, so that groups can assemble a genuinely comparative study. Attribute counts *include* the class attribute; all of these files ship with Weka (folder `$WEKA_HOME/data`).

3.3 Suggested Dataset Pairings

To make the required comparative analysis concrete, the following pairings are recommended (any two UCI datasets are acceptable, but the contrast must be justified):

- **Clinical screening:** `diabetes` + `heart-statlog` — same problem type, very different sample size.
- **Imbalance across domains:** `credit-g` (70:30) + `sick` (≈95:5) — how metric and sampling choice change with the imbalance ratio.
- **Nominal vs. numeric:** `mushroom` + `ionosphere` — filter choices diverge completely.
- **Small vs. large:** `glass` or `iris` + `letter` or `pendigits` — overfitting and runtime.
- **Missing-value stress test:** `hepatitis` + `arrhythmia` — imputation vs. row deletion at two scales.
- **High-dimensional:** `sonar` + `optdigits` — attribute selection and PCA before/after modeling.

3.4 Recommended Project Workflow

Figure 1 summarises the expected end-to-end workflow. Both datasets must pass through the same documented pipeline, and the pipeline must be re-run per dataset so that the comparison in Section 5 of the report is meaningful.

3.5 Examples of Problem Formulations

Examples of problem formulations include (but are not limited to):

- **Comparative classification:** train and compare at least four classifiers (e.g. `J48` (Quinlan, 1986), `RandomForest`, `Logistic`, `SMO`) on both datasets, and explain the differences in terms of dataset characteristics.
- **Effect of discretisation / normalisation:** quantify how `Discretize`, `Standardize` and `Normalize` change the ranking of models on numeric data (e.g. `optdigits`, `waveform-5000`).
- **Missing-value handling:** compare `ReplaceMissingValues` with row/attribute removal on `hepatitis`, `arrhythmia` or `vote` and report the effect on performance.
- **Feature selection:** filter-based (`InfoGainAttributeEval`+`Ranker`) vs. wrapper-based (`CfsSubsetEval`+`BestFirst`) vs. `PrincipalComponents` on `sonar`, `ionosphere` or `arrhythmia`.

Table 1: Recommended UCI datasets — supervised learning and numeric/high-dimensional data (all available as `.arff` in the standard Weka data folder). Verify the exact counts on the UCI dataset page before writing the report.

Dataset (ARFF)	**Inst.**	**Attr.**	**Type**	**Class.**	**Suggested use**
Binary classification — medical / clinical					
`diabetes`	768	8	num[†]	2	Medical screening; zero-coded missing values
`heart-statlog`	270	13	num	2	Small-sample learning; ROC/AUC analysis
`breast-w`	699	9	num	2	Trees, rules and k-NN comparison
`hepatitis`	155	19	mixed	2	Missing-value imputation study
Binary classification — credit / risk					
`credit-g`	1000	20	mixed	2	Class imbalance (70:30); cost-sensitive learning
`credit-a`	690	14	mixed	2	Mixed attribute types; discretisation effects
Imbalance and rare classes					
`sick`	3772	29	mixed	2	Severe imbalance ($\approx$5% positive); SMOTE
`hypothyroid`	3772	29	mixed	4	Multi-class with rare classes; metric choice
Multi-class / high-dimensional					
`iris`	150	4	num	3	Pipeline sanity check; clustering vs. labels
`glass`	214	9	num	6	Clustering; confusable class boundaries
`segment`	2310	19	num	7	Ensembles vs. single trees
`vowel`	990	13	mixed	11	Many-class problem; confusion analysis
`letter`	20000	16	num	26	Scalability; large multi-class
`optdigits`	5620	64	num	10	PCA vs. raw features; high dimension
`pendigits`	10992	16	num	10	k-NN / SMO timing and memory study
`sonar`	208	60	num	2	Curse of dimensionality; feature selection
`ionosphere`	351	34	num	2	Feature selection; noisy attributes
`arrhythmia`	452	279	num[†]	16	Extreme dimensionality + missingness
`waveform-5000`	5000	40	num	3	Noise robustness; ensemble comparison

[†]Missing values are encoded as `0` (or `?`); they must be handled explicitly.

Table 2: Recommended UCI datasets — nominal data, association analysis and rule learning.

Dataset (ARFF)	Inst.	Attr.	Type	Class.	Suggested use
Nominal / association analysis					
`mushroom`	8124	22	nom.	2	Apriori; nominal preprocessing
`vote`	435	16	nom.	2	Association rules with missing values
`soybean`	683	35	nom.	19	Many classes with few instances each
`zoo`	101	16	mixed	7	Clustering vs. ground-truth classes
`audiology`	226	69	nom.	24	Heavy missingness; many classes
Rule learning / small data					
`balance-scale`	625	4	num	3	`J48` vs. `OneR` vs. `JRip`
`kr-vs-kp`	3196	36	nom.	2	Rule learning; very accurate baselines

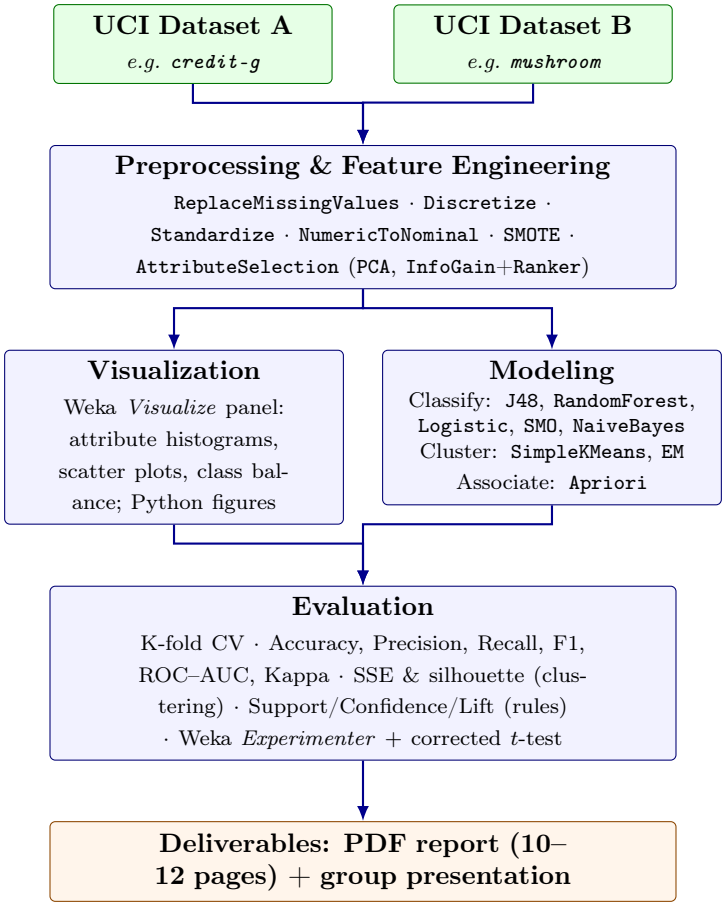


Figure 1: Recommended end-to-end workflow for the group project, applied to both selected UCI datasets.

Table 3: Minimum experimental scope for the group project.

Component	Minimum requirement
Datasets	$\geq$ 2 UCI datasets; $\geq$ 1 with $\geq$ 1,00 instances; $\geq$ 1 with missing/nominal attributes; the two must be contrasted explicitly.
Preprocessing	$\geq$ 3 documented filters, each with a stated purpose; $\geq$ 1 comparative experiment showing the effect of a preprocessing choice.
Visualization	$\geq$ 4 figures (attribute distributions, class balance, correlations/scatter, model output such as a tree plot or ROC curve), each interpreted in the text.
Modeling	$\geq$ 4 classification or regression algorithms *and* $\geq$ 1 unsupervised component (`SimpleKMeans`/`EM` *or* `Apriori`).
Evaluation	10-fold cross-validation (per dataset) *plus* one independent protocol (percentage split or repeated CV); report accuracy, precision, recall, F1, ROC–AUC and so on.
Comparison	A comparative table across models *and* across the two datasets, produced with the Weka *Experimenter* or an equivalent reproducible setup.

- **Clustering:** apply `SimpleKMeans` / `EM` to `iris`, `glass` or `zoo`; choose k using within-cluster SSE, then compare clusters with the true class labels (purity, confusion matrix).
- **Association analysis:** mine `Apriori` (Agrawal & Srikant, 1994) rules on `mushroom`, `vote` or `soybean`; tune support/confidence and interpret rules using support, confidence and lift.
- **Class imbalance:** compare plain learners with `SMOTE` (Chawla et al., 2002) (via `FilteredClassifier`), `CostSensitiveClassifier` and threshold adjustment on `credit-g` or `sick`.
- **Ensembles vs. single models:** `RandomForest` / `Bagging` / `AdaBoostM1` vs. `J48`, including a runtime and memory comparison.
- **Statistical comparison of algorithms:** use the Weka *Experimenter* over several datasets or repeated runs and test significance (corrected resampled t-test or Wilcoxon) (Demšar, 2006).

3.6 Minimum Experimental Requirements

Table 3 defines the minimum experimental scope. A project that does not meet every row is considered incomplete.

3.7 Recommended Weka Filters and Their Purposes

Choosing the *right* filter is part of the assessment. Table 4 maps common data problems to the appropriate Weka filter(s); groups are expected to justify why the chosen filter matches their data.

3.8 Notes

1. Apply the principles and techniques learned in the course via **Weka** (e.g., data preprocessing with filters, model training with classifiers/regressors, evaluation with cross-validation).

Table 4: Mapping data problems to Weka filters (package names abbreviated).

Purpose	**Recommended filter(s)**	**Notes / typical use**
Missing values	`ReplaceMissingValues`; `RemoveWithValues`; `RemoveUseless`	Mean/mode imputation vs. row removal; justify against missingness rate (compare `hepatitis` vs. `arrhythmia`).
Scaling	`Standardize`; `Normalize`	Required before distance-based methods: `SMO`, `IBk` (k-NN), `SimpleKMeans`, `Logistic`.
Discretisation	`Discretize`; `PKIDiscretize` (supervised)	Needed for `Apriori` and helpful for `NaiveBayes`; also a feature-engineering experiment.
Type conversion	`NumericToNominal`; `NominalToBinary`; `StringToWordVector`	`NumericToNominal` is mandatory before `Apriori`; `NominalToBinary` for logistic-type models.
Noise reduction	`RemoveDuplicates`; `InterquartileRange`; `RemovePercentage`	Outlier removal must be reported and justified; never applied to the test set alone.
Feature selection	`AttributeSelection` + `CfsSubsetEval`/`BestFirst` (wrapper); `InfoGainAttributeEval`/`Ranker` (filter)	Reduces dimensionality on `sonar`, `ionosphere`, `arrhythmia`; compare wrapper vs. filter cost.
Dimensionality reduction	`PrincipalComponents`	Applied before or after selection on `optdigits`; compare with raw attributes.
Class imbalance	`SMOTE`, `SpreadSubsample`, `Resample`	Wrap in `FilteredClassifier` so resampling happens *inside* each CV fold; use with `credit-g`, `sick`.
Instance resampling	`Resample`	Stratified train/test split; fixed random seed for reproducibility.
Feature construction	`AddExpression`; `MathExpression`; `Copy`	Derive ratios/indices; document the formula in the report.

Python (pandas, scikit-learn, matplotlib/seaborn) may be used for cleaning, feature engineering and visualization, but the modeling and evaluation workflow must be documented and reproducible.

2. Include the source of information. Provide clear references to the UCI dataset(s) used and any other resources that benefit the project, including licenses/usage constraints.
3. Use datasets that are reasonably large so that meaningful features can be extracted; ensure they can be handled in **Weka** (e.g., ARFF/CSV formats). If needed, convert UCI CSV files to ARFF and document the schema.
4. Avoid any form of data leakage: all preprocessing that uses class information (imputation by class mean, supervised discretisation, feature selection, SMOTE, normalisation) must be applied *inside* the cross-validation loop, e.g. by nesting filters inside `FilteredClassifier` or by using Weka's Knowledge Flow.

3 Project Report

Each group must submit a single, comprehensive project report (Weka-based) that both proposes and documents the entire project. The report should include:

1. Background and Objectives

- Background information on the topic and problem context.
- Clear statement of the project objectives and research questions.
- A short statement of *why* the two chosen UCI datasets were selected and what contrast they are expected to reveal.

2. Data Sources and Description

- Data sources and availability: the exact UCI dataset pages, UCI ID/DOI, original donor and year, and the license/usage constraints.
- Descriptive statistics (e.g., number of instances and features, class distribution, missing values).
- Attribute-type breakdown (numeric / nominal / binary) and, for both datasets, a short discussion of data quality issues (missing values, redundant attributes, class imbalance).
- Intended Weka data formats (ARFF/CSV) and any schema notes.

3. Methods and Workflow in Weka

- Preprocessing pipeline (e.g., `ReplaceMissingValues`, `Standardize`, `PrincipalComponents`); parameter settings and rationale.

- Modeling approaches (classifiers such as `J48`, `RandomForest`, `Logistic`; clustering with `SimpleKMeans`; association with `Apriori`); hyperparameters and feature selection.
- Evaluation protocols (e.g., train/validation split or 10-fold cross-validation), metrics (accuracy, precision, recall, F1, ROC/AUC; for clustering, silhouette or within-cluster SSE; for association, support/confidence/lift), and justification.
- Tools used within Weka (Explorer, Knowledge Flow, Experimenter) and how they were configured.
- For each of the two datasets, the pipeline must be described separately (or with an explicit table showing which steps differ), so that the comparison in Section 5 is traceable.

4. Tasks Performed

- Concrete tasks undertaken (data collection/cleaning, feature engineering, model training/tuning, validation, interpretation).
- Reasons for choosing specific preprocessing, visualization, and data mining methods in Weka.
- A division of labour for the 5–6 group members (who built which pipeline component, who ran which experiments, who prepared which figures).

5. Results and Analysis

- Quantitative results from Weka (confusion matrices, ROC curves, AUC, comparative tables from Experimenter).
- Comparative analysis of models and hyperparameters; ablation or sensitivity analysis where relevant.
- A direct cross-dataset comparison: which models/preprocessing steps generalise and which do not, and what property of each dataset explains the difference.
- Visualizations generated via Weka (e.g., decision tree plots, ROC curves) and their interpretation.

6. Discussion

- Key findings and insights relative to objectives.
- Limitations, data or modeling challenges, and threats to validity.
- Recommendations and potential future work.

7. References and Data Provenance

- Complete citations for datasets, tools, and literature.
- Provenance notes and any licenses/usage constraints.

Table 5: Individual contribution table (to be completed by the group and included in the report).

Member (ID)	Role	Specific contribution	Workload
	Dataset A owner	Data cleaning, schema, pipeline A	%
	Dataset B owner	Data cleaning, schema, pipeline B	%
	Modeling lead	Classifiers, tuning, Experimenter runs	%
	Visualization lead	All figures, presentation graphics	%
	Evaluation lead	CV protocol, metrics, statistics	%
	Writing/coordination	Report integration, references	%

8. Team Contributions

- A contribution table (Table 5) listing every member's specific contribution and the approximate workload share. **All** members must be able to explain the whole pipeline during the presentation.

4 Presentations

At the end of the semester, all members of each group are required to present their project. A rubric will be used to assess individual performance in the oral presentation. Demonstrating workflows and key results is encouraged.

Guidelines for a 5–6 member group:

- Total presentation time: 15 minutes, followed by 5 minutes of questions.
- Every member speaks for a minimum of 2–3 minutes; members who do not speak receive no mark for the presentation component.
- The group must give a Weka workflow (e.g. running a classifier on one of the UCI datasets, or replaying a saved Knowledge Flow), and show the cross-dataset comparison table.
- Slides are submitted by 9 November 2026 together with the report; file name `CDS535_Group_TeamName_Slides.pdf`.
- Each member should expect questions on the parts they own *and* on the overall design decisions.

5 Deadlines

Task	Deadline
Report submission	4 November, 2026
Presentation submission	9 November, 2026
Presentation	10 and 17 November, 2026

6 Assessment

Component	Weight
Report	20%
Presentation	10%
Total	**30%**

The Report component assesses the data-mining content (dataset selection and justification, preprocessing, visualization, modeling, evaluation, comparative analysis, reproducibility) as well as the quality of writing; the Presentation component assesses each member's individual understanding and communication.

References

Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules. *Proceedings of the 20th International Conference on Very Large Data Bases (VLDB)*, 487–499.

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, *16*, 321–357. https://doi.org/10.1613/jair.953

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, *7*, 1–30.

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Quinlan, J. R. (1986). Induction of decision trees. *Machine Learning*, *1*(1), 81–106. https://doi.org/10.1007/BF00116251

Schlimmer, J. (1987). Mushroom [Dataset]. https://doi.org/10.24432/C5959T

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